

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - SHORT ANSWER TEST

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 - Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
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Section / Batch:	2023	Date:	04/09/2026

Code of Conduct

I, N. Deepika / 192312196 (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: deepika

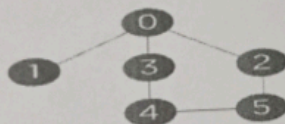
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A – Short Answer Test

- Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop (), 3) Pop (), 3) Pop Construct AVL tree for the following elements. Mention the
- Given a collection of elements 10,20,30,40,50,60,70,80. The Key element to be search 70. Compare how searching is performed in Linear Search and Binary Search. Identify the efficient search method. Justify.
- What is the postfix and prefix forms of the following expression? $A+B*(C-D)/(P-R)$
- Construct AVL tree for the following elements. Mention the Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10
- Construct a heap by inserting 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45 into a binary heap.
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Perform the merge sort for the following inputs. Also, show the result for each step of iteration. 40,20,70,14,60,61,97,30.
- How the following data set can be sorted by quick sort. Show each step-in sorting. 78,23,31,88,43,55,67,55.
- Perform Open Hashing & Linear Probing for the given set of 0,1,81, 4,64,25,16,36, 9 and 49.
- Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completion	10	Completes all questions accurately within the allotted time	Completes most questions on time	Requires additional time	Unable to complete the test

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

perform the following operations using stack. Assume the size of the stack is 5 and have a value of invert 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operation.

- 1) Invert the elements in the stack.
- 2) pop()
- 3) pop()
- 3) pop construct AVL tree for the following elements. Mention the.

Initial stack from position 0 to 4:

0	1	2	3	4
22	55	33	66	88

Top = 88

- 1) After inversion: 88, 66, 33, 55, 22

Now Top = 22

- 2) pop removes 22 Remaining: 88, 66, 33, 55
- 3) pop removes 55 Remaining: 88, 66, 33.

final answer: 88, 66, 33.

- 2) Given a collection of elements 10, 20, 30, 40, 50, 60, 70, 80. The key elements to be search 70. compare how searching is performing in linear search and Binary search. Identify the efficient search method. justify.

elements: 10, 20, 30, 40, 50, 60, 70, 80.

search key = 70

Linear search

checks one by one: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 70$

comparisons = 7.

Binary search

step search range

middle

comparison

1

10-80

40

$70 > 40$

2

50-80

60

$70 > 60$

3

70-80

70

found.

comparisons = 3.

Efficient method:-

Binary Search is more efficient because:

Linear search: $O(n)$

Binary search: $O(\log n)$.

Binary search requires the data to be sorted.

3) what is the post-fix and prefix forms of the following expression? $A + B * (C - D) / [P - R]$.

expression:

$A + B * (C - D) / (P - R)$

prefix:

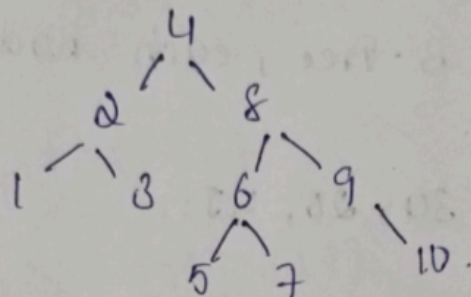
$+ A / * B - C D - P R$

post-fix:

$- A B C D - * P R - / +$

construct AVL tree for the following elements.
Mention the rotations in each step of iteration.
1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Insert: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

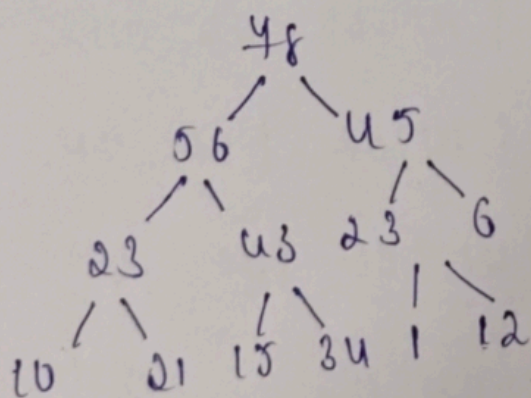


All balance factors are within $-1, 0, +1$.

5) construct ~~an~~ tree heap by inserting 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45 into a binary heap.

Elements: 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45

Assuming a max heap:



Array representation:-

[78, 56, 45, 23, 43, 23, 6, 10, 21, 15, 34, 1, 12]

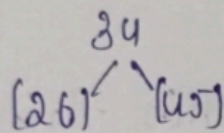
6) Construct a B-tree of order 3 by inserting numbers: 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95.

elements:

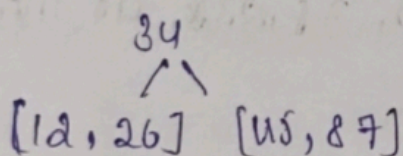
34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95.

for an order-3 B-tree, each node can contain at most 2 keys.

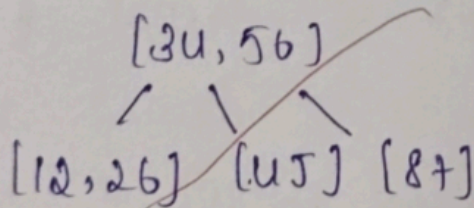
After inserting 34, 26, 45:



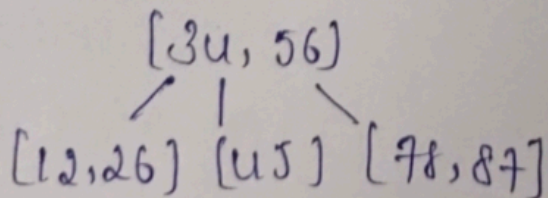
After inserting 12, 87:



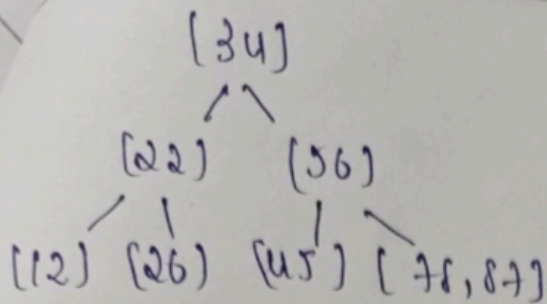
After inserting 56:



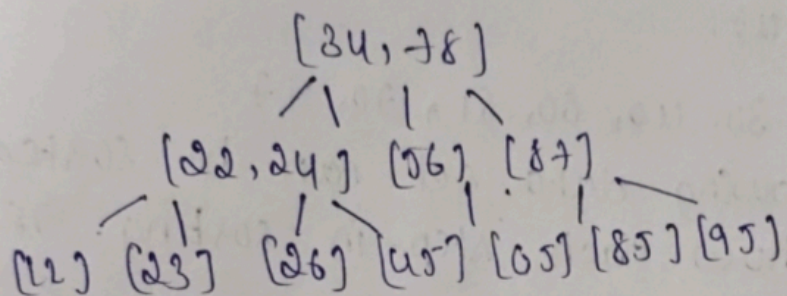
After inserting 78:



After inserting 22:



After inserting 24, 65, 85 and 95, the final B-Tree is:



7) perform the merge sort, for the following input
Also, show the result for each step of iteration.
on. 40, 20, 70, 14, 60, 61, 97, 30.

Input: 40, 20, 70, 14, 60, 61, 97, 30.

Divide.

40 20 70 14 | 60 61 97 30

40 20 | 70 14 | 60 61 | 97 30

40 | 20 | 70 | 14 | 60 | 61 | 97 | 30.

Merge:

20 40

14 70

60 61

30 97.

merge again:-

14 20 40 70
30 60 81 97

final merge:

14 20 30 40 60 61 70 97.

sorted output:

14, 20, 30, 40, 60, 61, 70, 97.

- 8) How the following data set can be sorted by quick sort show each step in sorting. 78, 23, 31, 88, 43, 55, 67, 55.

Input: 78, 23, 31, 88, 43, 55, 67, 55.

using the last element as pivot.

first pivot = 55.

After partition: 21, 31, 43, 55, 55, 88, 67, 78.

Left part: 23 31 43.

Right part: 88 67 78.

sort left part

pivot = 43:

23 31 43.

pivot = 31;

23 31.

23 31 43
sorting right part

88 67 78

pivot = 78:

67 78 88

final sorted array:

23 31 43 55 67 78 88

9) perform open hashing & linear probing for the given set of 0, 1, 81, 4, 64, 25, 16, 36, 9 and 49.

Given:

0, 1, 81, 4, 64, 25, 16, 36, 9, 49.

Assume:

Hash function: $h(k) = k \bmod 10$.

A) open hashing | separate chaining.

Index values.

0	0
1	1 → 81
2	-
3	-
4	4 → 64
5	25
6	16 → 36
7	-

Because:

$$81 \bmod 10 = 1$$

$$64 \bmod 10 = 4$$

$$36 \bmod 10 = 6$$

$$49 \bmod 10 = 9$$

B) Linear probing

Index value

0 0

1 1

2 81

3 49

4 4

5 64

6 25

7 16

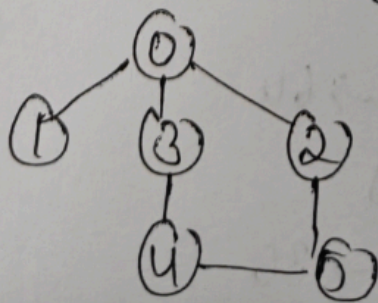
8 36

9 9

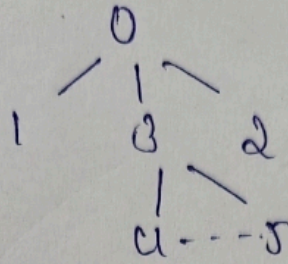
final linear probing array:

[0, 1, 81, 49, 4, 64, 25, 16, 36, 9]

10) Consider the following graph & construct-BFS.



from the graph, starting node is 0.
connections:



starting from 0:

BFS steps

visit 0: Queue [0]

visit 1, 3, 2: [1, 3, 2].

visit 1: No new node

visit 3: Add 4.

[2, 4].

visit 2: Queue [4, 5].

visit 4: No new node.

visit 5: No new node.

BFS traversal:

0 → 1 → 3 → 2 → 4 → 5.

Wrong